Transfer learning

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One complete transfer-mechanics record.

Keep givens, fitting rule, validation results and representation limits together.

Keep one habit: Recall → W1–W3/W5 → independent extractor → project record → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Similarity and normalisation. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Sequence representations. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Pooling and spatial information. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- A source model may learn useful features before we have many labels for a new target task. **Transfer learning** reuses those learned features or parameters. One approach freezes the **feature extractor** and fits a new **output head**.

  Fine tuning instead updates some or all of the reused parameters, usually together with the head.

- Use a transparent stand-in for a source-trained extractor: f(x) = ReLU(x1 + x2), with supplied weights [1, 1] and bias 0. ReLU keeps a positive value and replaces a negative value with zero.

  Here x1 and x2 are nonnegative fictional event counts, so f(x) is their sum. We did not train this extractor on a source dataset; this exercise demonstrates the mechanics of reuse, not measured transfer benefits.

- The target is a high-activity label. The extractor returns a single number h, not a class label or probability. A new head converts h into the target **prediction**.

Similarity and normalisation

> **My remembered idea, example and correction**

Sequence representations

> **My remembered idea, example and correction**

Pooling and spatial information

> **My remembered idea, example and correction**

Lesson reference for workbook practice

Part	Operation	During this exercise
Input	[count of type 1, count of type 2]	Keep order and units
Extractor	h = max(0, x1 + x2)	Freeze weights and bias
New head	Predict 1 if h >= threshold	Fit the threshold on target training data

---

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main example: supplied frozen h = ReLU(x1 + x2), weights [1, 1], bias 0. Training T1 [1, 1] label 0; T2 [1, 3] label 0; T3 [4, 2] label 1; T4 [5, 3] label 1. Features 2, 4, 6, 8. Candidate thresholds 3, 5, 7 make 1, 0, 1 mistakes; choose 5. Predict class 1 at equality. Baseline fits the training majority; two labels of each class tie, so choose class 0.

  Validation V1 [2, 1] label 0 gives h 3 and prediction 0; V2 [3, 4] label 1 gives h 7 and prediction 1. Head 2/2, baseline 1/2 on the same rows. New inference [2, 3] gives h 5 and class 1. Picture instead uses h = left + 2 × right, training features 1, 4, 7 and threshold 5; query [2, 2] predicts 1 without a supplied reference label. Changed query [2, 1] predicts 0.

  New-target U [1, 3] label 0 and V [3, 1] label 1 both collapse to h 4 under the original extractor. A deterministic head cannot separate them. Supplied repair ReLU(x1 – x2) with threshold 1 separates these two rows but was not fitted. This demonstration has no source pretraining, full fine-tuning, from-scratch timing comparison or final evaluation.

- For **input** [4, 2], compute h using the original extractor and the class at threshold 5. Identify the frozen weights and bias, the fitted quantity, and why h is not a probability.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main example: supplied frozen h = ReLU(x1 + x2), weights [1, 1], bias 0. Training T1 [1, 1] label 0; T2 [1, 3] label 0; T3 [4, 2] label 1; T4 [5, 3] label 1. Features 2, 4, 6, 8. Candidate thresholds 3, 5, 7 make 1, 0, 1 mistakes; choose 5. Predict class 1 at equality. Baseline fits the training majority; two labels of each class tie, so choose class 0.

  Validation V1 [2, 1] label 0 gives h 3 and prediction 0; V2 [3, 4] label 1 gives h 7 and prediction 1. Head 2/2, baseline 1/2 on the same rows. New inference [2, 3] gives h 5 and class 1. Picture instead uses h = left + 2 × right, training features 1, 4, 7 and threshold 5; query [2, 2] predicts 1 without a supplied reference label. Changed query [2, 1] predicts 0.

  New-target U [1, 3] label 0 and V [3, 1] label 1 both collapse to h 4 under the original extractor. A deterministic head cannot separate them. Supplied repair ReLU(x1 – x2) with threshold 1 separates these two rows but was not fitted. This demonstration has no source pretraining, full fine-tuning, from-scratch timing comparison or final evaluation.

- For the four target **training** records, give the number of mistakes at thresholds 3, 5 and 7 and choose the fitted threshold. Then score that head and the declared constant baseline on V1 and V2.

  Explain which data fitted the head.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main example: supplied frozen h = ReLU(x1 + x2), weights [1, 1], bias 0. Training T1 [1, 1] label 0; T2 [1, 3] label 0; T3 [4, 2] label 1; T4 [5, 3] label 1. Features 2, 4, 6, 8. Candidate thresholds 3, 5, 7 make 1, 0, 1 mistakes; choose 5. Predict class 1 at equality. Baseline fits the training majority; two labels of each class tie, so choose class 0.

  Validation V1 [2, 1] label 0 gives h 3 and prediction 0; V2 [3, 4] label 1 gives h 7 and prediction 1. Head 2/2, baseline 1/2 on the same rows. New inference [2, 3] gives h 5 and class 1. Picture instead uses h = left + 2 × right, training features 1, 4, 7 and threshold 5; query [2, 2] predicts 1 without a supplied reference label. Changed query [2, 1] predicts 0.

  New-target U [1, 3] label 0 and V [3, 1] label 1 both collapse to h 4 under the original extractor. A deterministic head cannot separate them. Supplied repair ReLU(x1 – x2) with threshold 1 separates these two rows but was not fitted. This demonstration has no source pretraining, full fine-tuning, from-scratch timing comparison or final evaluation.

- For the first-count-greater target, explain why the original extractor cannot distinguish U [1, 3] from V [3, 1]. Trace the hand-chosen [1, -1] extractor and threshold 1.

  Does this show that **fine tuning** has been trained and independently validated?

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main example: supplied frozen h = ReLU(x1 + x2), weights [1, 1], bias 0. Training T1 [1, 1] label 0; T2 [1, 3] label 0; T3 [4, 2] label 1; T4 [5, 3] label 1. Features 2, 4, 6, 8. Candidate thresholds 3, 5, 7 make 1, 0, 1 mistakes; choose 5. Predict class 1 at equality. Baseline fits the training majority; two labels of each class tie, so choose class 0.

  Validation V1 [2, 1] label 0 gives h 3 and prediction 0; V2 [3, 4] label 1 gives h 7 and prediction 1. Head 2/2, baseline 1/2 on the same rows. New inference [2, 3] gives h 5 and class 1. Picture instead uses h = left + 2 × right, training features 1, 4, 7 and threshold 5; query [2, 2] predicts 1 without a supplied reference label. Changed query [2, 1] predicts 0.

  New-target U [1, 3] label 0 and V [3, 1] label 1 both collapse to h 4 under the original extractor. A deterministic head cannot separate them. Supplied repair ReLU(x1 – x2) with threshold 1 separates these two rows but was not fitted. This demonstration has no source pretraining, full fine-tuning, from-scratch timing comparison or final evaluation.

- A report omits the source checkpoint and pre**process**ing, claims faster learning without timing, and repeatedly chooses settings using the final evaluation set.

  Name the missing records and propose a fair comparison with a model trained from scratch.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A separate frozen extractor is h = 2*x1 + x2. **Training input**s [1, 0], [1, 2], [2, 1], [3, 1] have labels 0, 0, 1, 1.

  Compare thresholds 2, 4 and 6 using **prediction** 1 when h >= threshold; choose the fewest mistakes, breaking ties toward the smaller threshold. Score it on validation [2, 0] with label 1 and [0, 2] with label 0.

  Compare with the **training**-majority baseline, whose tie rule chooses 0.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Fit a new head independently

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Add a labelled **transfer learning evidence** sheet to Image-pattern Network Investigation. Show how you can trace a reused **feature extractor** and a newly fitted **output head**.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Fit and compare for **Transfer learning**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Transfer learning**. Proposed explanation: "Frozen means the whole model cannot learn". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A separate frozen extractor is h = 2*x1 + x2. **Training input**s [1, 0], [1, 2], [2, 1], [3, 1] have labels 0, 0, 1, 1.

  Compare thresholds 2, 4 and 6 using **prediction** 1 when h >= threshold; choose the fewest mistakes, breaking ties toward the smaller threshold. Score it on validation [2, 0] with label 1 and [0, 2] with label 0.

  Compare with the **training**-majority baseline, whose tie rule chooses 0.

> **My reasoning and evidence**

▼ Extra practice · Pictured case

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Transfer a fixed **feature extractor** while identifying what is actually refitted.

- Frozen extractor h =left+2right. New-task **training input**s [1, 0]label 0,[2, 1]label 0,[1, 3]label 1. Class 1 when h >=threshold; candidates 3 or5.

- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

- Which part was trained for the new task? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-5/28

Transfer learning

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Extract features

Training h values are 1, 4, 7.



2. Fit new head

Threshold 3 makes 1 mistake; threshold 5 makes 0, so choose 5.



3. Check held-out input

Query [2, 2]has h = 6, so the head predicts class 1; its reference is not supplied.

CHANGE ONE THING

Query changes to [2, 1].

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ Reference · Project milestone

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Transfer a fixed **feature extractor** while identifying what is actually refitted.
- Which labelled step matches this detail? Query [2, 2]has h = 6, so the head predicts class 1; its reference is not supplied.
- 1. Check held-out **input**
- 2. Extract features
- 3. Fit new head
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Check held-out **input** Query [2, 2]has h = 6, so the head predicts class 1; its reference is not supplied.

Transfer learning can keep a representation fixed while **fitting** a smaller **output head** for a new task. Here the extractor always applies the same weights, while new labels choose the head's threshold.

Applying the chosen head to a query is inference. State which part was frozen and which was fitted; this arithmetic does not establish the extractor is useful for every task.

Changed-case question and optional hint

- Changed-case practice: Query changes to [2, 1]. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. Its h becomes 4 and the fitted head predicts class 0.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Its h becomes 4 and the fitted head predicts class 0. Extractor stays fixed; the changed **input** crosses the chosen head's boundary.

Repair and retain the record

Review Similarity and normalisation (ai-5/27). Goal: Calculate vector length and normalise a nonzero vector. Short practice: Why do the original two metrics select different winners?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Sequence representations (ai-5/25). Goal: Encode an ordered sequence with an explicit vocabulary. Short practice: Which record lets the program distinguish the first and last zeros?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Pooling and spatial information (ai-5/22). Goal: Compute max and mean pooling under an explicit window rule. Short practice: Can the max map reveal where the 4 originally appeared?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | | |
> |---|---|
> | | |
> | | |
> | | |
> | | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

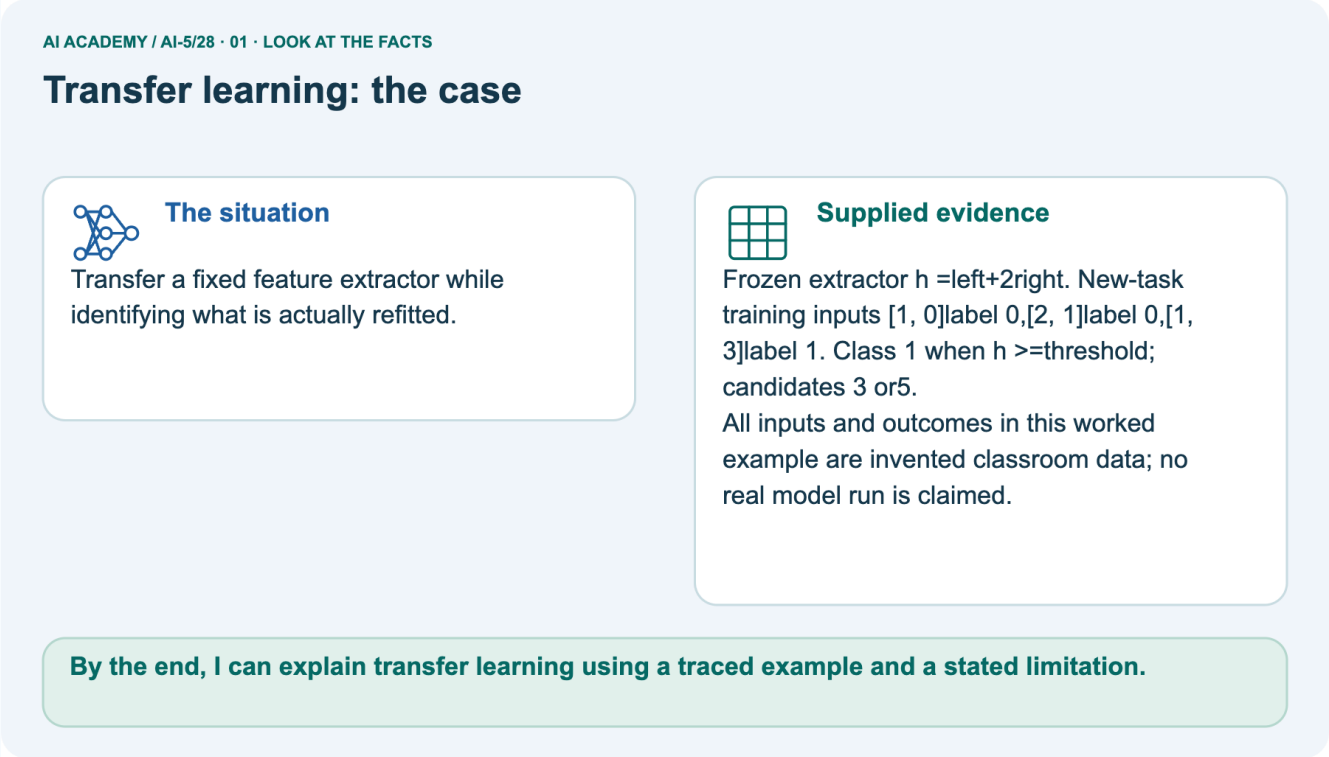
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Transfer learning: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

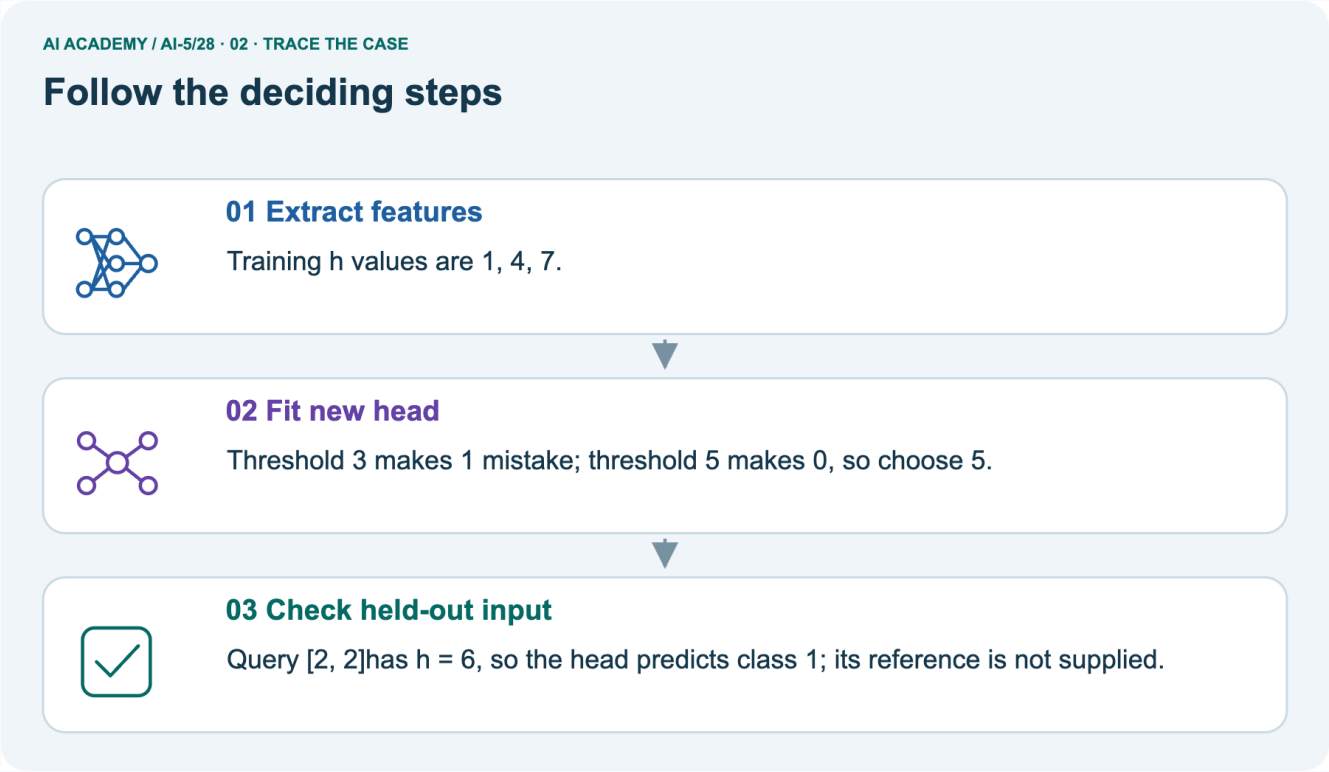
Transfer a fixed **feature extractor** while identifying what is actually refitted. Supplied facts: Frozen extractor h =left+2right. New-task **training input**s [1, 0]label 0,[2, 1]label 0,[1, 3]label 1.

Class 1 when h >=threshold; candidates 3 or5. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



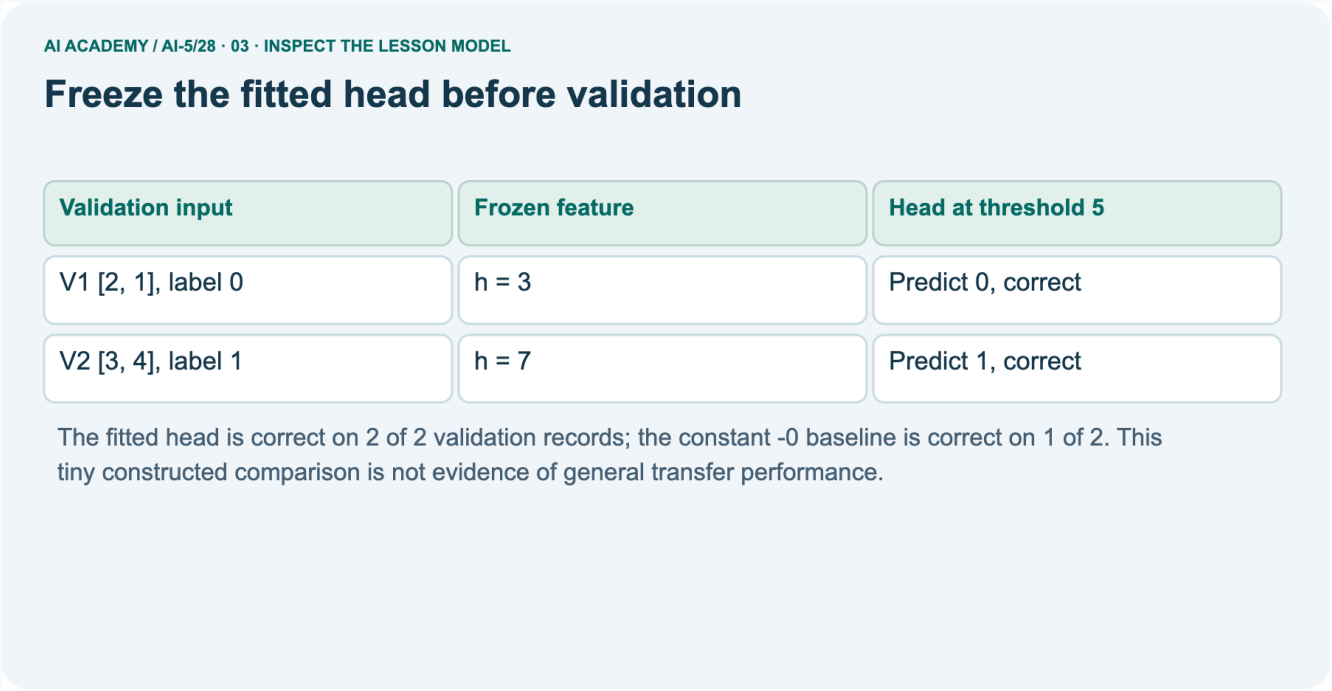
Follow the labelled picture with the matching explanation. Open the vector diagram

Extract features: **Training** h values are 1, 4, 7. → Fit new head: Threshold 3 makes 1 mistake; threshold 5 makes 0, so choose 5. → Check held-out **input**: Query [2, 2]has h = 6, so the head predicts class 1; its reference is not supplied.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Freeze the fitted head before validation



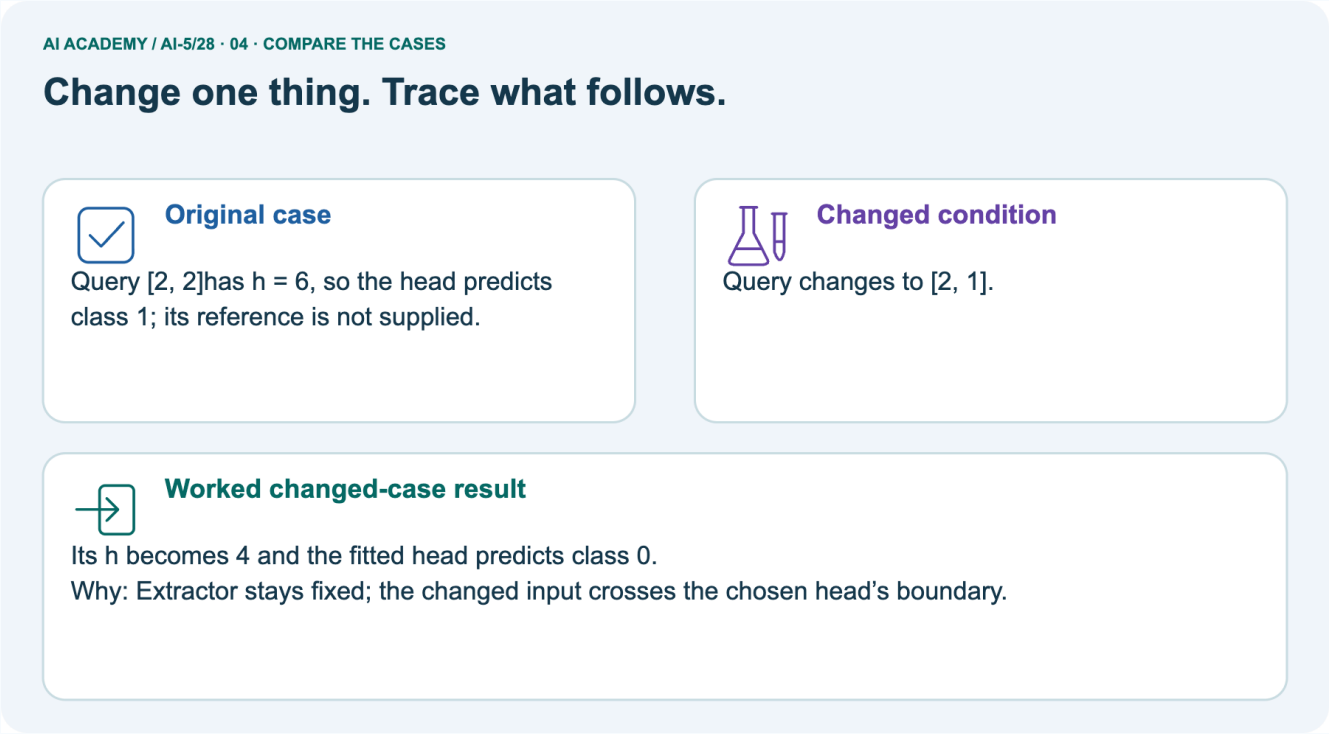
Follow the labelled picture with the matching explanation. Open the vector diagram

Freeze the fitted head before validation. V1 [2, 1], label 0 | h = 3 | Predict 0, correct. V2 [3, 4], label 1 | h = 7 | Predict 1, correct

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

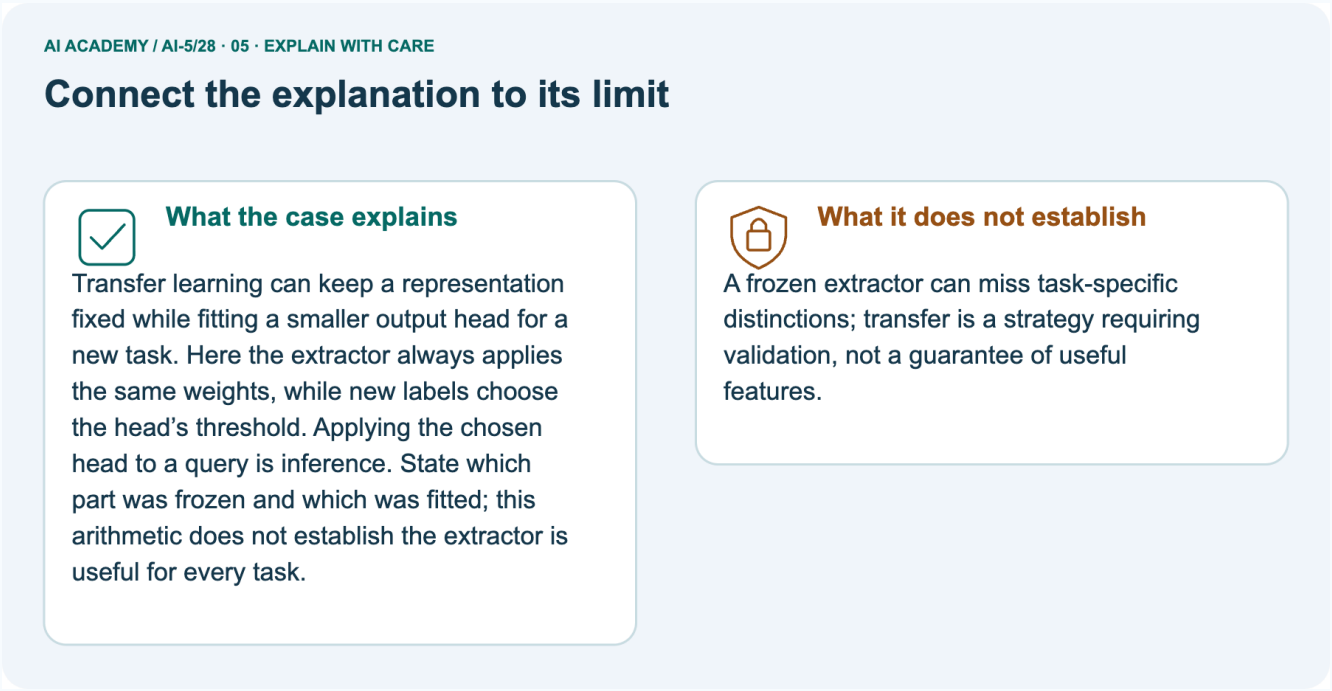
Original case: Query [2, 2]has h = 6, so the head predicts class 1; its reference is not supplied. Changed condition: Query changes to [2, 1]. Expected result: Its h becomes 4 and the fitted head predicts class 0.

Reason: Extractor stays fixed; the changed **input** crosses the chosen head’s boundary.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Transfer learning can keep a representation fixed while **fitting** a smaller **output head** for a new task. Here the extractor always applies the same weights, while new labels choose the head's threshold.

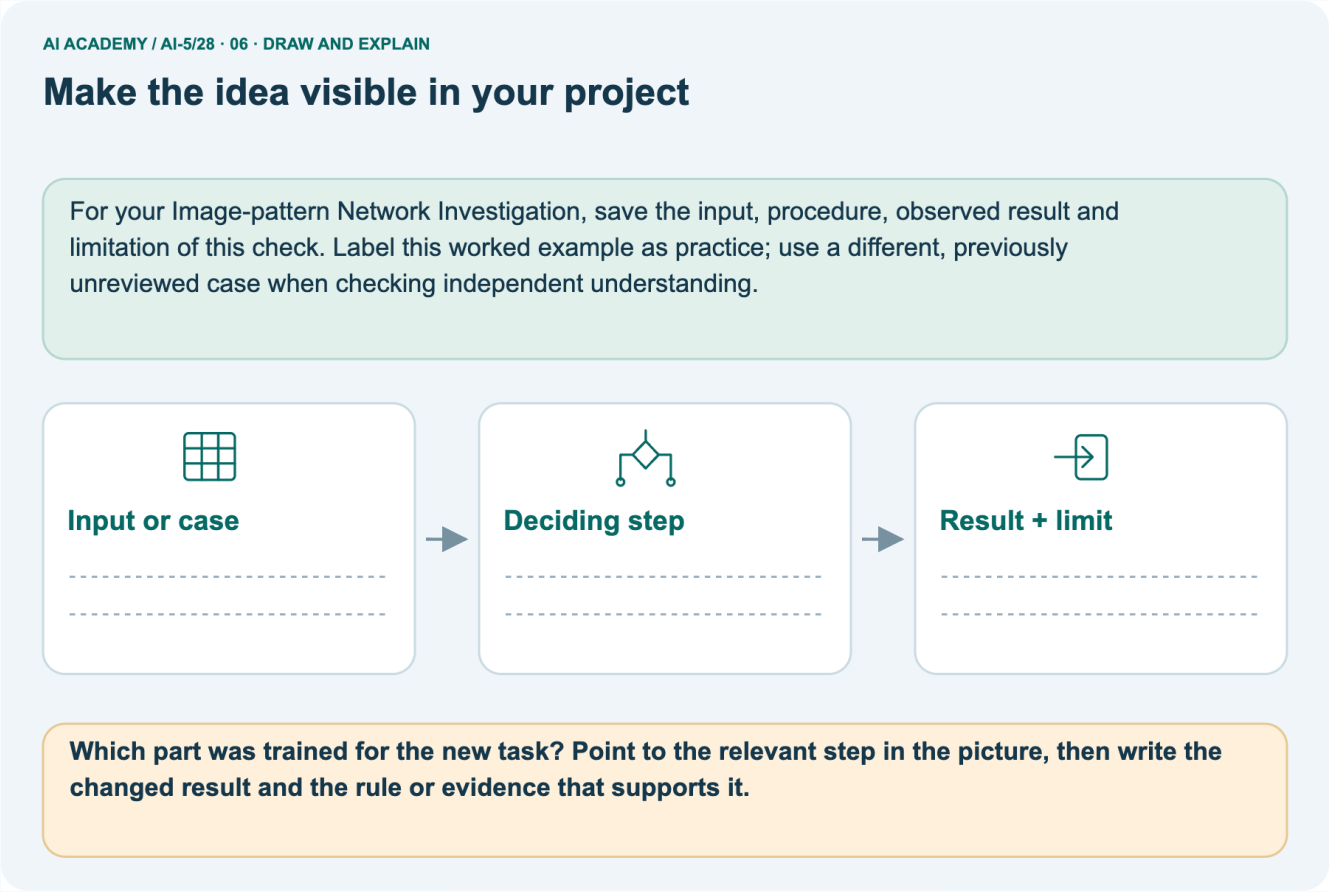
Applying the chosen head to a query is inference. State which part was frozen and which was fitted; this arithmetic does not establish the extractor is useful for every task.

Limit: A frozen extractor can miss task-specific distinctions; transfer is a strategy requiring validation, not a guarantee of useful features.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which part was trained for the new task? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

01. Transfer learning: the case

02. Follow the deciding steps

03. Freeze the fitted head before validation

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project
</details>

<details>
<summary>Optional interactive practice and completed work</summary>

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.
</details>